Enter to Learn, Leave to Succeed

Project Title	Beyond Buy and Hold: An AI-Powered Integrated FA+TA Decision Engine for Nifty IT Stocks
AI Case Title	Multi-Agent AI Underwriting Co-Pilot for Automated Term Insurance Risk Assessment and Fraud Detection
Working Prototype	Local Python pipelines + HTML "CIP Decision Dashboards" (Equity Research – FA+TA tab and Underwriting Co-Pilot tab); screenshots in Section 3
Host Organisation	HDFC Life Insurance Company Limited
Role	Equity Research and Wealth Management Intern
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Corporate Mentor	Mr. Saurav Shrivastav
Faculty Guide	Prof. Indrajit Lahiri

Section 1

AI Problem Fitment

1.1 What is the problem? Who does it affect, and how?

Two information-heavy, repetitive decision processes surfaced during the internship in the Equity Research and Wealth Management function. First, comparing all ten Nifty IT Index constituents on both fundamental strength (PE, PEG, EBIT margin, ROE, FCF yield, revenue growth) and technical positioning (trend, momentum, volatility indicators) is done manually today, stock by stock, which is slow and leaves room for inconsistent judgement calls between analysts. Second, on the underwriting side of a life-insurer, verifying that an applicant's KYC, income, and medical documents are internally consistent - and flagging cases that look like misrepresentation - is a repetitive, document-heavy check that is currently done by a human reviewer for every single case, regardless of how routine it is.

Both problems affect three groups in similar ways: the analyst or underwriter (time spent on repetitive screening instead of judgement-heavy work), the end customer or investor (slower turnaround), and the organisation (operational efficiency and consistency of decisions). Both are specific, structured, and rule-expressible, which makes them well suited to AI-assisted automation rather than full manual replacement.

1.2 How is it currently solved without AI?

Fundamental and technical analysis of the Nifty IT basket is currently maintained as a manual peer-comparison spreadsheet, refreshed periodically, with technical charts checked separately on a charting platform - there is no single, automatically refreshed view that combines both. Underwriting document checks are done fully manually: every KYC, income, and medical document is read and cross-checked by a human reviewer, with no automated first-pass consistency check to separate routine files from ones that need closer attention.

1.3 Why is AI viable here?

Both tasks are strong fits for AI-assisted automation because the inputs are structured (financial ratios, price/volume series, applicant document fields) and the scoring logic is transparent and rule-expressible - the AI layer does not invent the rules, it applies them consistently and then reasons over the result in plain language. In the FA+TA engine, a deterministic Python scoring function computes the fundamental score and a rule-based function computes the technical signal; an LLM only fuses the two into a recommendation and an explanation, which keeps the process auditable. In the underwriting co-pilot, each of the five agents produces structured, inspectable JSON rather than an opaque final verdict, and every recommendation still requires human sign-off before a policy decision is made.

1.4 How the problem was confirmed as real

Both problems are well-documented, recognised operational themes: sell-side and buy-side equity research widely combines fundamental and technical screens rather than relying on either alone, and underwriting turnaround time and consistency on borderline cases are widely acknowledged challenges in the life-insurance industry. Both also surfaced directly in my own work during the internship - the peer-comparison workbook and Power BI/Excel outputs used for the equity research deliverables, and the underwriting-efficiency theme that came up while analysing insurer performance.

Section 2

AI Solution Design and Build

2.1 What did you build?

I built two working automation pipelines that share the same architecture pattern - deterministic, transparent scoring in Python, with an LLM used only for structured fusion and plain-language explanation - plus a single HTML dashboard with two tabs to review both.

A. FA+TA Decision Engine (core project): For each of the Nifty IT Index constituents, the pipeline (1) reads the fundamental peer-comparison workbook and computes a transparent 0–100 fundamental score from PEG ratio, EBIT margin, ROE, FCF yield, and revenue growth; (2) pulls one year of daily price history via the Yahoo Finance API and computes ten technical indicators - SMA-20, SMA-50, EMA-20, RSI-14, MACD line/signal/histogram, Bollinger Bands, Stochastic %K, ATR-14, and OBV - and converts them into a rule-based Bullish/Bearish/Neutral signal; (3) sends both structured outputs to an LLM (Groq, llama-3.3-70b-versatile) with a fixed prompt template that returns only a JSON verdict of Buy/Hold/Avoid, a signal_tension flag (true when fundamentals and technicals disagree), and a 2–3 sentence, number-citing justification; and (4) exports the consolidated results to Excel for review in Power BI / the dashboard.

B. Multi-Agent AI Underwriting Co-Pilot (AI case): Five sequential agents - KYC Verification, Income Verification, Medical Assessment, Fraud Detection, and Decision Support - each call the same LLM with a JSON-only structured prompt. The KYC agent checks name/DOB consistency; the Income agent computes an income-consistency ratio (declared vs. payslip-implied annual income) and flags it if outside 0.85–1.15; the Medical agent maps health data to a Standard/Sub-Standard/Decline risk class; the Fraud agent cross-references the three upstream outputs for mismatches; and the Decision Support agent consolidates everything into a 0–100 risk score and one of four recommendations (Accept Standard Terms / Accept with Loading / Refer to Manual Underwriter / Decline), always for a human underwriter to review.

Both pipelines write to a single “CIP Decision Dashboards” HTML page with an Equity Research – FA+TA tab and an Underwriting Co-Pilot tab, so results can be reviewed visually instead of only in raw Excel/JSON (see Section 3).

2.2 Tools / platforms used

- Python (pandas, numpy, yfinance, openpyxl) for data pull, scoring, and Excel export
- Groq API (llama-3.3-70b-versatile) for structured JSON signal-fusion and agent reasoning, called with response_format=json_object and temperature=0.2 for consistency

- A custom HTML/CSS/JS dashboard ("CIP Decision Dashboards") that reads the exported .xlsx and .json files directly in the browser
- Planned, documented (not yet wired) extensions: n8n for daily scheduling of the FA+TA script, Power BI for a live refreshed dashboard, and Zapier/Make.com for signal-tension email alerts

2.3 Data / documents used

FA+TA engine: real, public market data - daily OHLCV price history from Yahoo Finance for each Nifty IT constituent - combined with my own fundamental peer-comparison workbook (Fundamental_Analysis_IT_sector.xlsx). Underwriting co-pilot: synthetic, illustrative applicant data only. Two synthetic cases were used - one fully consistent case and one with a deliberately seeded income inconsistency - to exercise the Fraud Detection agent. No real customer KYC, financial, or medical documents were accessed at any stage; this is the responsible ceiling given the confidentiality and regulatory sensitivity of real underwriting data.

2.4 Output produced

FA+TA Decision Engine - recommendation table (fundamental score, technical signal, Buy/Hold/Avoid, signal tension flag, and justification) for each company, exported to FA_TA_Decision_Output.xlsx. Of the eleven Nifty IT constituents in the fundamental workbook, eight completed the full pipeline in this run and are summarised below; LTIMindtree, OFSS, and Hexaware did not return a technical signal in this run (ticker/price-history fetch issue - see Section 2.5) and are a follow-up item.

6 BUY	2 HOLD	0 AVOID	0 SIGNAL TENSION

FA+TA Decision Engine - aggregate recommendation counts across the 8 companies covered in this run.

Company	Fund. Score	Technical Signal	Recommendation	Signal Tension
TCS	82.9	Neutral	Hold	No
Infosys	79.6	Neutral	Buy	No
HCL Tech	73.1	Bullish	Buy	No
Wipro	64.7	Bullish	Buy	No

Company	Fund. Score	Technical Signal	Recommendation	Signal Tension
Tech Mahindra	60.1	Bullish	Buy	No
Persistent Systems	68.3	Neutral	Hold	No
Coforge	63.4	Neutral	Buy	No
Mphasis	67.1	Bullish	Buy	No

Table 2.1 - FA+TA Decision Engine output (fundamental_score shown is the calculated composite score).

Underwriting Co-Pilot - for each synthetic case, a full audit trail across all five agents plus a final risk score and recommendation:

2	1	40	1
CASES PROCESSED	FRAUD FLAGGED	AVG RISK SCORE	HUMAN OVERRIDE

Underwriting Co-Pilot - aggregate outcomes across the two synthetic test cases.

Case ID	Income Flag	Fraud Flag	Risk Score	Recommendation
SYN-001 (clean)	Consistent (ratio 1.00)	false (Low)	20 / 100	Accept Standard Terms
SYN-002 (seeded fraud)	Inconsistent (ratio 2.46)	true (Medium)	60 / 100	Refer to Manual Underwriter

Table 2.2 - Underwriting Co-Pilot output for the two synthetic test cases (underwriting_copilot_results.json).

2.5 Limitations

- FA+TA recommendations are only as good as the peer-comparison workbook and the point-in-time price pull - they are a screening aid, not investment advice, and were not back-tested against forward returns in this run.
- Three of the eleven Nifty IT constituents (LTIMindtree, OFSS, Hexaware) did not return a technical signal in this run and need a ticker/data-source fix before the next refresh.
- The underwriting co-pilot is not predictive - it is rule-guided LLM reasoning tested on two synthetic cases, not trained or validated on real claims/actuarial outcomes.

- Both pipelines depend on a single third-party LLM provider (Groq) with no fallback model configured yet, and neither has a persisted audit log of every run beyond the exported Excel/JSON files.
- Governance: an LLM-generated recommendation in either domain - investment or underwriting - raises real explainability, bias, and regulatory-compliance questions that a proof of concept does not resolve on its own.

Section 3

AI Demonstration and Use Evidence

3.1 Deployment level achieved

Proof of concept - no organizational deployment level claimed. Both pipelines were built, run end-to-end, and reviewed by me on the dashboard; neither was demonstrated to a supervisor or colleague during the internship, so I make no Level 1–3 organizational-adoption claim, only a working, self-tested proof of concept.

3.2 How the solution was demonstrated / tested

I ran the FA+TA engine end-to-end across the Nifty IT constituents and cross-checked the exported Excel output against the dashboard's aggregate counts (Buy/Hold/Avoid and signal-tension totals). I ran the underwriting co-pilot on both synthetic cases - one clean and one with a seeded income inconsistency - and confirmed the Fraud Detection agent correctly flagged the seeded case while leaving the clean case unflagged, with the Decision Support agent correctly escalating the flagged case to “Refer to Manual Underwriter” and requiring human override.

3.3 Who saw or used the solution

Both pipelines were built, run, and tested by me. They have not yet been demonstrated to or used by a supervisor or colleague. The screenshots below show the live local dashboard displaying both pipelines' outputs, followed by the raw terminal execution logs as first-hand proof that both scripts run end-to-end and export correctly.

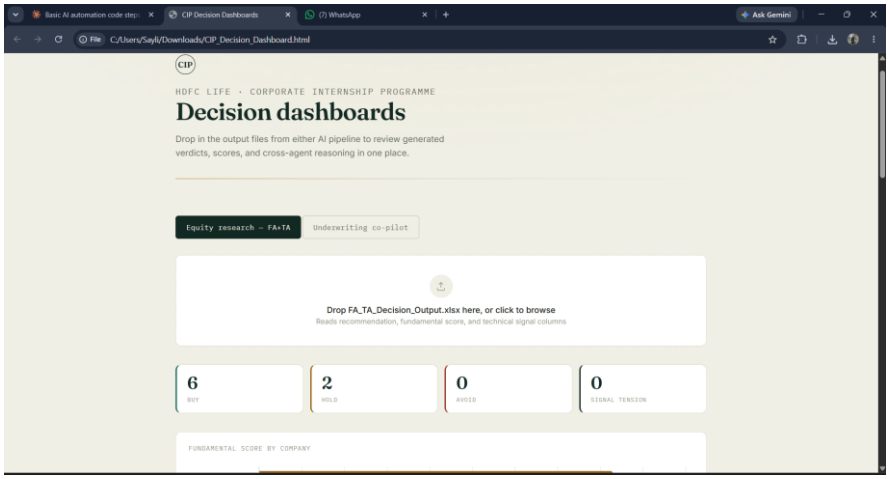


Figure P1 - Equity Research (FA+TA) dashboard tab: aggregate Buy/Hold/Avoid/Signal-Tension counts and per-company fundamental score, generated from FA_TA_Decision_Output.xlsx.

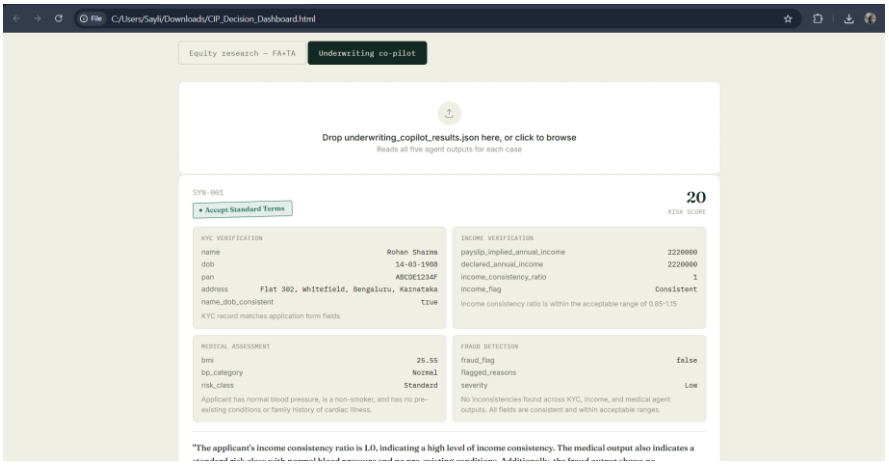


Figure P2 - Underwriting Co-Pilot dashboard tab: case SYN-001 showing all five agent outputs (KYC, Income, Medical, Fraud) and the final risk score / recommendation, generated from underwriting_copilot_results.json.

3.4 Terminal execution evidence

The two screenshots below are unedited PowerShell captures of the actual runs: the first shows fa_ta_decision_engine.py processing all eleven Nifty IT tickers end-to-end and exporting FA_TA_Decision_Output.xlsx, immediately followed by underwriting_copilot.py running its five-agent pipeline on case SYN-001; the second shows the raw underwriting_copilot_results.json content for case SYN-001 being re-read from disk to confirm it was saved correctly.

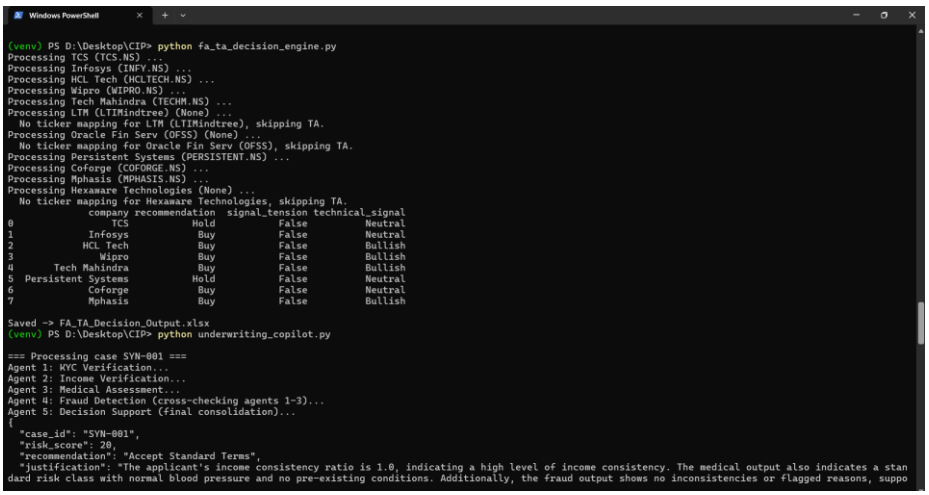


Figure P3 - Terminal log: fa_ta_decision_engine.py processing all 11 tickers (3 skipped for a ticker-mapping issue, as noted in Section 2.5), exporting FA_TA_Decision_Output.xlsx, then underwriting_copilot.py beginning its 5-agent run on case SYN-001.

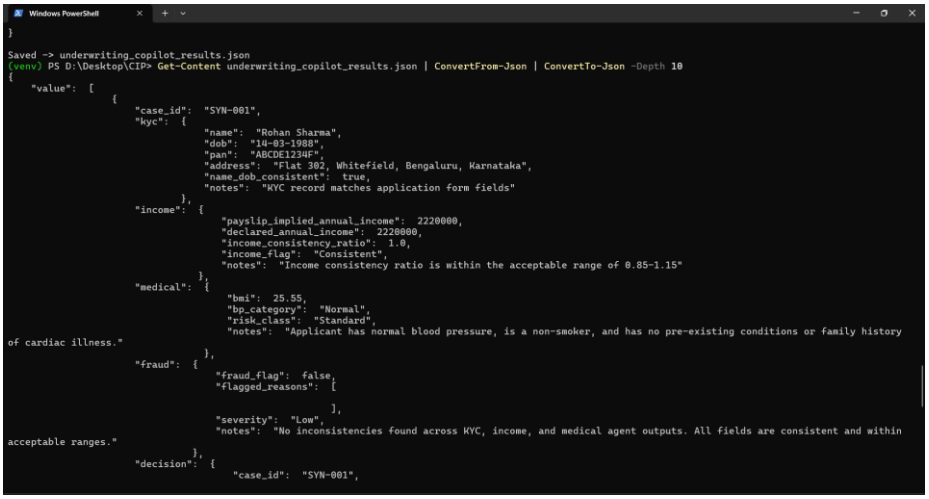


Figure P4 - Terminal log: underwriting_copilot_results.json read back from disk for case SYN-001, confirming the KYC, income, medical, and fraud agent outputs were saved exactly as produced.

Section 4

Organizational Response and Constraint

4.1 Why the work stopped at proof of concept

What I reached. Both pipelines were built and validated by me - the FA+TA engine on live market data across the Nifty IT basket, and the underwriting co-pilot on two synthetic cases. Neither was demonstrated to a supervisor or colleague, so I make no organizational deployment claim for either.

The data and mandate constraint. A production FA+TA tool would need sign-off from a SEBI-registered research analyst before any Buy/Hold/Avoid output could be shared externally, and a production underwriting tool only becomes useful once it runs on real applicant data - health records, financial documents, and prior-claim histories - which is confidential and regulated under IRDAI norms and India's data-protection requirements. As an intern in Equity Research and Wealth Management, I had neither the mandate nor the access to place either tool in front of the underwriting team or issue live recommendations, so the responsible ceiling for this internship was a working proof of concept.

4.2 What broader deployment would require

- Data permissions and access - formal sign-off to use governed historical price/fundamental data and, for underwriting, anonymised historical case data to validate against real outcomes.
- Regulatory and compliance clearance - SEBI research-analyst sign-off for any published FA+TA recommendation, and an IRDAI compliance review before any real applicant data touches the underwriting pipeline.
- Independent validation - back-testing of the FA+TA scoring logic against forward returns, and actuarial/fairness validation of the underwriting risk scores.
- Model governance and audit trail - versioned prompts, a documented fallback model, and a full log of every AI recommendation and human override.
- IT integration - a scheduled n8n/Power BI refresh for the FA+TA engine, and integration with the policy-administration system for the underwriting co-pilot, rather than standalone scripts and a local dashboard.
- People, training, and ownership - a named workflow owner and supervisor buy-in for a controlled pilot with full human oversight on both sides.

4.3 Organizational engagement and observation

No organizational respondent engaged with either prototype during the internship. Because both were kept at proof-of-concept stage and were not demonstrated to my corporate mentor or any colleague, there is no first-hand organizational observation to record here. I have chosen to state this plainly rather than attribute views to anyone who did not see the tools.

4.4 What I would do differently and how I would take it forward

- Fix the three missing tickers (LTIMindtree, OFSS, Hexaware) and add a retry/fallback data source before the next FA+TA refresh.
- Add a persisted, append-only run log for both pipelines, so every recommendation and every human override is auditable by design rather than only visible in the latest Excel/JSON export.
- Wire the already-documented n8n schedule and Power BI refresh for the FA+TA engine, and a signal-tension Zapier email alert, so the dashboard reflects a live rather than a one-off run.
- Stress-test the underwriting co-pilot with a larger, messier synthetic document set (including OCR-style noise) before proposing it for even a Level-1 demonstration.
- Lead with compliance in any future pitch - SEBI research-analyst sign-off for FA+TA, IRDAI data governance for underwriting - since that, not the model logic, is what would decide whether either idea can move beyond a proof of concept.

Section 5

Final AI Case Write-Up and Proof

This section confirms that the full case (Sections 1–5) is complete and internally consistent, lists the proof items attached, and carries the student declaration.

5.1 Proof items attached

#	Proof item	Format	File name / description
1	FA+TA Decision Engine source code	Python (.py)	fa_ta_decision_engine.py - fundamental scoring, 10-indicator technical analysis, Groq LLM fusion
2	Underwriting Co-Pilot source code	Python (.py)	underwriting_copilot.py - 5-agent pipeline (KYC, Income, Medical, Fraud, Decision Support)
3	FA+TA pipeline output	Excel (.xlsx)	FA_TA_Decision_Output.xlsx - recommendation, fundamental score, technical signal per company
4	Underwriting pipeline output	JSON (.json)	underwriting_copilot_results.json - full 5-agent output for both synthetic cases
5	Terminal execution logs	Annotated screenshots	Figures P3–P4 - live PowerShell run of both scripts end-to-end, confirming the pipelines execute and export correctly
6	Dashboard screenshots	Annotated screenshots	Figures P1–P2 - the Equity Research and Underwriting Co-Pilot dashboard tabs
7	Documented workflow	This document	Sections 1–5 (problem fitment, build, demonstration, constraints, proof)

5.2 Final consistency check across problem, solution, deployment, response, and proof

Element	What the case states	Consistent?
Problem	Manual FA+TA screening across 10 Nifty IT constituents and manual underwriting document cross-checking are both slow and inconsistent (Sec. 1).	Yes
Solution	Two automated pipelines - an FA+TA scoring/LLM-fusion engine and a 5-agent underwriting co-pilot - matching the stated problems (Sec. 2).	Yes
Deployment	Working proof of concept, run end-to-end on real market data (FA+TA) and synthetic applicant data (Underwriting); no organizational level claimed (Sec. 3).	Yes
Response	No organizational demonstration was carried out at this stage, so no organizational response is claimed; reasons stated candidly (Sec. 4).	Yes
Proof	Source code, Excel/JSON outputs, terminal execution logs, and dashboard screenshots (Sec. 5.1) evidence exactly the above - all first-hand.	Yes
Data honesty	Underwriting used synthetic data only; FA+TA used public market data; no real customer KYC/financial/medical data was accessed at any stage.	Yes

Student Declaration

I confirm that the problems and solutions described relate to genuine operational themes identified during my internship. The FA+TA Decision Engine was run on real, publicly available market data; the Underwriting Co-Pilot was built and tested by me on synthetic, illustrative data only, and no real customer KYC, financial, or medical data was accessed at any stage. I have not claimed that HDFC Life Insurance Company Limited uses either workflow. Both deliverables are working proofs of concept that I built and tested myself; I have not claimed any demonstration to, use by, or adoption within the organization, and no organizational response is reported because none took place. All sections of this bundle are internally consistent, and the proof items listed in Section 5.1 are attached.

Sayali (25PGDMFM052)

Student Signature		Date: 06.07.2026